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Linux Administration
4/**1**/26

Server Setup **II**

Before doing anything in the server environment for Ubuntu and CentOS you should upgrade the packages with the following command…

Sudo apt upgrade- Upgrades the different software packages to the latest version in the Ubuntu server environment.

Ubuntu server:

1. **cd /etc/apt** - Changes directory to the APT configuration folder. Commands written on **3/25/26.**

cat sources.list - Brings you to the sources file in the Ubuntu server environment.

```
## distribution.
deb http://us.archive.ubuntu.com/ubuntu/ jammy-updates main restricted
# deb-src http://us.archive.ubuntu.com/ubuntu/ jammy-updates main restricted

## N.B. software from this repository is ENTIRELY UNSUPPORTED by the Ubuntu
## team. Also, please note that software in universe WILL NOT receive any
## review or updates from the Ubuntu security team.
deb http://us.archive.ubuntu.com/ubuntu/ jammy universe
# deb-src http://us.archive.ubuntu.com/ubuntu/ jammy universe
deb http://us.archive.ubuntu.com/ubuntu/ jammy-updates universe
# deb-src http://us.archive.ubuntu.com/ubuntu/ jammy-updates universe

## N.B. software from this repository is ENTIRELY UNSUPPORTED by the Ubuntu
## team, and may not be under a free licence. Please satisfy yourself as to
## your rights to use the software. Also, please note that software in
## multiverse WILL NOT receive any review or updates from the Ubuntu
## security team.
deb http://us.archive.ubuntu.com/ubuntu/ jammy multiverse
# deb-src http://us.archive.ubuntu.com/ubuntu/ jammy multiverse
deb http://us.archive.ubuntu.com/ubuntu/ jammy-updates multiverse
# deb-src http://us.archive.ubuntu.com/ubuntu/ jammy-updates multiverse

## N.B. software from this repository may not have been tested as
## extensively as that contained in the main release, although it includes
## newer versions of some applications which may provide useful features.
## Also, please note that software in backports WILL NOT receive any review
## or updates from the Ubuntu security team.
deb http://us.archive.ubuntu.com/ubuntu/ jammy-backports main restricted universe multiverse
# deb-src http://us.archive.ubuntu.com/ubuntu/ jammy-backports main restricted universe multiverse

deb http://security.ubuntu.com/ubuntu/ jammy-security main restricted
# deb-src http://security.ubuntu.com/ubuntu/ jammy-security main restricted
deb http://security.ubuntu.com/ubuntu/ jammy-security universe
# deb-src http://security.ubuntu.com/ubuntu/ jammy-security universe
deb http://security.ubuntu.com/ubuntu/ jammy-security multiverse
# deb-src http://security.ubuntu.com/ubuntu/ jammy-security multiverse
aavbox26@Ubuntu1:/etc/apt$ _
```

Sudo add-apt-repository multiverse -Adds a **repository** to add different sources to the Ubuntu server environment. Multiverse is software that is not free, cannot be supported but can be added. Software in multiverse cannot be supported and does not get any type of updates.

Repository added on **3/30/26.**

```
aavbox26@Ubuntu1:/$ sudo add-apt-repository multiverse
[sudo] password for aavbox26:
Adding component(s) 'multiverse' to all repositories.
Press [ENTER] to continue or Ctrl-c to cancel.
Hit:1 http://us.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://us.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://us.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:5 https://ppa.launchpadcontent.net/git-core/ppa/ubuntu jammy InRelease
Hit:6 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu jammy InRelease
Reading package lists... Done
aavbox26@Ubuntu1:/$
```

Sudo add-apt-repository ppa:git-core/ppa and **Sudo add-apt-repository ppa:graphics-drivers/ppa -** Adds two different **PPA** to the server environment. After adding the PPA to the server, to see that the PPA was added we would write the following command....

ls /etc/apt/sources.list.d/.

We then see a line that shows the different **PPA** within the directory for the server. The PPA installs were completed on 3/26/26.

```
If you run into old documentation referring to other PPAs, you can help us by consolidating referenc
es to this PPA.

If someone wants to go ahead and start prototyping on `software-properties-gtk` on what the GUI shou
ld look like, please start hacking!

## Help us Help You!

We use the donation funds to get the developers hardware to test and upload these drivers, please co
nsider donating to the "community" slider on the donation page if you're loving this PPA:

http://www.ubuntu.com/download/desktop/contribute
More info: https://launchpad.net/~graphics-drivers/+archive/ubuntu/ppa
Adding repository.
Press [ENTER] to continue or Ctrl-c to cancel.
Adding deb entry to /etc/apt/sources.list.d/graphics-drivers-ubuntu-ppa-jammy.list
Adding disabled deb-src entry to /etc/apt/sources.list.d/graphics-drivers-ubuntu-ppa-jammy.list
Adding key to /etc/apt/trusted.gpg.d/graphics-drivers-ubuntu-ppa.gpg with fingerprint 2388FF3BE10A76
F638F80723FCAE110B1118213C
Hit:1 http://us.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://us.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://us.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:4 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:5 https://ppa.launchpadcontent.net/git-core/ppa/ubuntu jammy InRelease
Get:6 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu jammy InRelease [24.3 kB]
Get:7 http://us.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3,329 kB]
Get:8 http://us.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1,258 kB]
Get:9 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu jammy/main amd64 Packages [28.6 k
B]
Get:10 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu jammy/main Translation-en [6,076
 B]
Fetched 5,030 kB in 2s (2,446 kB/s)
Reading package lists... Done
aavbox26@Ubuntu1:/etc/apt$ ls /etc/apt/sources.list.d/
git-core-ubuntu-ppa-jammy.list  graphics-drivers-ubuntu-ppa-jammy.list
aavbox26@Ubuntu1:/etc/apt$
```

Sudo apt update- Updating packages command. The update command was done on 3/26/26.

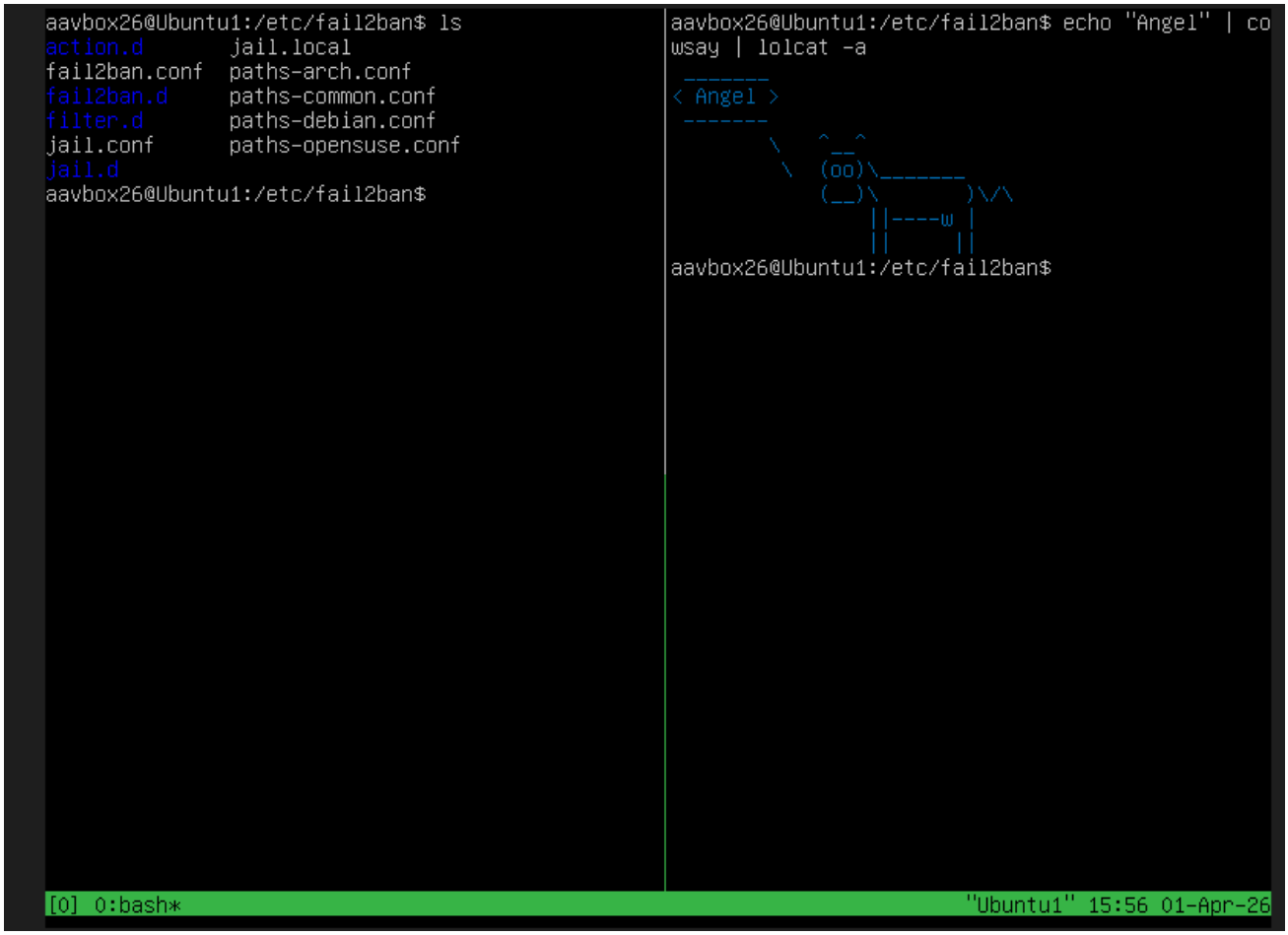
```
aavbox26@Ubuntu1:/etc/apt$ sudo apt update
Hit:1 http://us.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://us.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://us.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Hit:5 https://ppa.launchpadcontent.net/git-core/ppa/ubuntu jammy InRelease
Hit:6 https://ppa.launchpadcontent.net/graphics-drivers/ppa/ubuntu jammy InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
73 packages can be upgraded. Run 'apt list --upgradable' to see them.
aavbox26@Ubuntu1:/etc/apt$ _
```

2. **Sudo apt install tmux-** Checks to see if tmux is installed. Tmux splits the terminal. Tmux was installed by default. Installed on **3/25/26.** Finished installs for emacs and fail2ban on **3/26/26.**

```
aavbox26@Ubuntu1:/$ sudo install tmux
install: missing destination file operand after 'tmux'
Try 'install --help' for more information.
aavbox26@Ubuntu1:/$ sudo apt install tmux
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
tmux is already the newest version (3.2a-4ubuntu0.2).
0 upgraded, 0 newly installed, 0 to remove and 63 not upgraded.
aavbox26@Ubuntu1:/$
```

Install <u>command</u> for Tmux.

The steps to get into the tmux program, **tmux** is a program that splits windows into multiple terminals. The way I started the **tmux** program within the ubuntu server admin terminal was writing **tmux.** Then I did **ctrl b**, then **shift 5,** to split the screen vertically, these two commands open a new terminal screen within the server.



To close the tmux program we write **Tmux kill-session,** this command ends the session that splits the two terminal screens.

Sudo apt install emacs , sudo apt install fail2ban- A command that installs the emacs text editor and the fail2ban prevention software for the Ubuntu server.

cd /etc/fail2ban - This command shows that the **fail2ban** is its own directory. Remember to add a space in front of cat to ensure the command works, without the space there will be errors or complete command failure.

cd /etc/emacs- This command shows that **emacs** is its own directory. Remember to add a space in front of cat to ensure the command works, without the space there will be errors or complete command failure.

When writing **ls** you can see the files that are within the directories **fail2ban** and **emacs.** Emacs does not contain a directory with files.

Sudo nano jail.conf- Jail.conf is one of the files in fail2ban with this command we can open the file to see its contents. When opening the file you can see the different security default settings for **fail2ban.**

Sudo cp jail.conf jail.local- A command that makes a copy of the jail.conf file and renames it to jail.local. Writing **ls** we can see that the original file and the copied file are now within the same directory in **"fail2ban".**

```
  GNU nano 6.2                                    jail.conf
#
# WARNING: heavily refactored in 0.9.0 release.  Please review and
#          customize settings for your setup.
#
# Changes:  in most of the cases you should not modify this
#           file, but provide customizations in jail.local file,
#           or separate .conf files under jail.d/ directory, e.g.:
#
# HOW TO ACTIVATE JAILS:
#
# YOU SHOULD NOT MODIFY THIS FILE.
#
# It will probably be overwritten or improved in a distribution update.
#
# Provide customizations in a jail.local file or a jail.d/customisation.local.
# For example to change the default bantime for all jails and to enable the
# ssh-iptables jail the following (uncommented) would appear in the .local file.
# See man 5 jail.conf for details.
#
# [DEFAULT]
# bantime = 1h
#
# [sshd]
# enabled = true
#
# See jail.conf(5) man page for more information



# Comments: use '#' for comment lines and ';' (following a space) for inline comments


[INCLUDES]
                                        [ Read 968 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo
```

Screenshot of **fail2ban original settings** and file in **nano above.**

```
aavbox26@Ubuntu1:/$ cd /etc/fail2ban
aavbox26@Ubuntu1:/etc/fail2ban$ ls
action.d       fail2ban.d  jail.conf  jail.local       paths-common.conf  paths-opensuse.conf
fail2ban.conf  filter.d    jail.d     paths-arch.conf  paths-debian.conf
aavbox26@Ubuntu1:/etc/fail2ban$ _
```

Screenshot showing the original and copied file of **jail.conf.**

```
aavbox26@Ubuntu1:/etc/fail2ban$ sudo systemctl status fail2ban.service
● fail2ban.service - Fail2Ban Service
     Loaded: loaded (/lib/systemd/system/fail2ban.service; disabled; vendor preset: enabled)
     Active: inactive (dead)
       Docs: man:fail2ban(1)
aavbox26@Ubuntu1:/etc/fail2ban$
```

The original command to see the status of **fail2ban** is **sudo systemctl status fail2ban.service.**

To get **fail2ban** to work we write **sudo systemctl enable fail2ban.service.** Then **sudo systemctl start fail2ban.service,** to start fail2ban.

Finally we write **sudo systemctl status fail2ban.service,** to see the current status of fail2ban and see if it is running properly.

```
aavbox26@Ubuntu1:/etc/fail2ban$ sudo systemctl start fail2ban.service
aavbox26@Ubuntu1:/etc/fail2ban$ sudo systemctl status fail2ban.service
● fail2ban.service - Fail2Ban Service
     Loaded: loaded (/lib/systemd/system/fail2ban.service; enabled; vendor preset: enabled)
     Active: active (running) since Fri 2026-03-27 02:59:34 UTC; 19s ago
       Docs: man:fail2ban(1)
   Main PID: 4627 (fail2ban-server)
      Tasks: 5 (limit: 4555)
     Memory: 12.0M
        CPU: 250ms
     CGroup: /system.slice/fail2ban.service
             └─4627 /usr/bin/python3 /usr/bin/fail2ban-server -xf start

Mar 27 02:59:34 Ubuntu1 systemd[1]: Started Fail2Ban Service.
Mar 27 02:59:35 Ubuntu1 fail2ban-server[4627]: Server ready
aavbox26@Ubuntu1:/etc/fail2ban$ _
```

Above is a screenshot showing **fail2ban working** in the **Ubuntu** server environment.

To open emacs within the Ubuntu server I first changed the directory by writing **cd /etc/emacs.** But there was an error with this so I wrote cd .. twice and changed the directory back to home and, then I wrote the command **emacs -nw**, which is emacs space No Window. No window works within the terminal space, instead of a separate graphics window.

To close the emacs new window you would press **ctrl x and then ctrl c,** this is very important to do as if you do not do this you will be stuck in the emacs welcome screen and would have to restart your server.

```
File Edit Options Buffers Tools Help
Welcome to GNU Emacs, one component of the GNU/Linux operating system.
To follow a link, click Mouse-1 on it, or move to it and type RET.
To quit a partially entered command, type Control-g.

Important Help menu items:
Emacs Tutorial          Learn basic Emacs keystroke commands
Read the Emacs Manual   View the Emacs manual using Info
(Non)Warranty           GNU Emacs comes with ABSOLUTELY NO WARRANTY
Copying Conditions      Conditions for redistributing and changing Emacs
More Manuals / Ordering Manuals  How to order printed manuals from the FSF

Useful tasks:
Visit New File          Specify a new file's name, to edit the file
Open Home Directory     Open your home directory, to operate on its files
Customize Startup       Change initialization settings including this screen

GNU Emacs 27.1 (build 1, x86_64-pc-linux-gnu, GTK+ Version 3.24.33, cairo version 1.16.0)
 of 2024-09-17, modified by Debian
Copyright (C) 2020 Free Software Foundation, Inc.

-UUU:%%--F1  *GNU Emacs*    All L19    (Fundamental) ----------------------------------------
End of buffer
```

3. **Sudo install apt lolcat-** Installs the lolcat software. Installed on 3/25/26.

Echo "Angel" | **cowsay** | **lolcat -a :** This is a command that prints text for the cowsay character to say. With lolcat there are different colors displayed in the terminal for the character and text in the Ubuntu server environment.

```
aavbox26@Ubuntu1:/etc/fail2ban$ echo "Angel" | cowsay | lolcat -a
 _______
< Angel >
 -------
        \   ^__^
         \  (oo)\_______
            (__)\       )\/\
                ||----w |
                ||     ||
aavbox26@Ubuntu1:/etc/fail2ban$
```

usr/share/cowsay/cows- use **- F-. - a** command to see different character options.

echo "hey" | **cowsay -f tux** | **lolcat -a** A command for a different character option like the linux Tux penguin in different colors and animated in the Ubuntu server environment.

```
aavbox26@Ubuntu1:/etc/fail2ban$ echo "hey" | cowsay -f tux | lolcat -a
 _____
< hey >
 -----
   \
    \
        .--.
       |o_o |
       |:_/ |
      //   \ \
     (|     | )
    /'\_   _/`\
    \___)=(___/

aavbox26@Ubuntu1:/etc/fail2ban$
```

4. **Sudo apt install vim**- installs vim on the ubuntu server. Installed on 3/25/26.

   Cd ~ - changes directory.

   Nano ~/.bashrc- opens a nano file in the ubuntu server environment.

A colorful text file is then opened, at the end of the file we write, **alias vi=vim** to change **vi to vim** in the terminal.

```
  GNU nano 6.2                    /home/aavbox26/.bashrc
# colored GCC warnings and errors
#export GCC_COLORS='error=01;31:warning=01;35:note=01;36:caret=01;32:locus=01:quote=01'

# some more ls aliases
alias ll='ls -alF'
alias la='ls -A'
alias l='ls -CF'

# Add an "alert" alias for long running commands.  Use like so:
#   sleep 10; alert
alias alert='notify-send --urgency=low -i "$([ $? = 0 ] && echo terminal || echo error)" "$(history>

# Alias definitions.
# You may want to put all your additions into a separate file like
# ~/.bash_aliases, instead of adding them here directly.
# See /usr/share/doc/bash-doc/examples in the bash-doc package.

if [ -f ~/.bash_aliases ]; then
    . ~/.bash_aliases
fi

# enable programmable completion features (you don't need to enable
# this, if it's already enabled in /etc/bash.bashrc and /etc/profile
# sources /etc/bash.bashrc).
if ! shopt -oq posix; then
  if [ -f /usr/share/bash-completion/bash_completion ]; then
    . /usr/share/bash-completion/bash_completion
  elif [ -f /etc/bash_completion ]; then
    . /etc/bash_completion
  fi
fi
alias vi=vim

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^_ Go To Line M-E Redo
```

We then write **ctrl O** to save/write out, then **press enter then ctrl x** to close the file.

Then in the ubuntu terminal we write **vi** and results of **vi=vim** are displayed.

We then write **:q** to **escape/exit** the screen. **- End of Question 4.**

Ubuntu server **Sources:**

https://linuxize.com/post/how-to-add-apt-repository-in-ubuntu/ **Shows quick command references for Ubuntu server. Question 1 source.** /etc/apt/sources.list.d/ **command source.**

https://www.scribd.com/document/803731589/4-Module-3-Installing-and-Exploring-cowsay-on-Ubuntu **Steps on how to install cowsay and examples. Question 3 source.**

https://www.tecmint.com/lolcat-color-output-linux-terminal/ **How to install lolcat with command examples and screenshots. Question 3 source.**

https://launchpad.net/~git-core/+archive/ubuntu/ppa **PPA example command.**

https://ubuntu.com/server/docs/how-to/software/package-management/ **Install and manage packages official documentation by Ubuntu.**

https://help.ubuntu.com/community/Fail2ban **Fail2ban configuration, install commands.**

https://www.digitalocean.com/community/tutorials/how-to-protect-ssh-with-fail2ban-on-ubuntu-20-04 **Installing Fail2ban command and setup etc.**

https://www.digitalocean.com/community/tutorials/how-to-use-the-emacs-editor-in-linux **How to install emacs.**

https://superuser.com/questions/203298/ubuntu-keep-emacs-in-the-terminal **How to enter the emacs screen.**

https://web.physics.ucsb.edu/~pcs/apps/editors/emacs-basics.html **How to exit emacs and other commands.**

https://linuxize.com/post/getting-started-with-tmux/ **Tmux example commands.**

Tutoring Assistance.

CentOS server:

cd /etc/yum- Changes directory to the YUM configuration folder. Command written on **3/30/26.**

cd /etc/yum.repos.d/- Shows the sources for CentOS, the listed sources in CentOS. Brings you to the directory of CentOS in YUM. Written on **3/30/36.**

Ls- ls the command that shows you a list of repositories within the **CentOS** server and **YUM.** Written on **3/30/36.**

```
[Angel@localhost yum]$ /etc/yum.repos.d/
-bash: /etc/yum.repos.d/: Is a directory
[Angel@localhost yum]$ cd /etc/yum.repos.d/
[Angel@localhost yum.repos.d]$ ls
CentOS-Linux-AppStream.repo          CentOS-Linux-FastTrack.repo
CentOS-Linux-BaseOS.repo             CentOS-Linux-HighAvailability.repo
CentOS-Linux-ContinuousRelease.repo  CentOS-Linux-Media.repo
CentOS-Linux-Debuginfo.repo          CentOS-Linux-Plus.repo
CentOS-Linux-Devel.repo              CentOS-Linux-PowerTools.repo
CentOS-Linux-Extras.repo             CentOS-Linux-Sources.repo
[Angel@localhost yum.repos.d]$
```

There were issues for the user and for root to get into the repository. With the following commands I was able to install the official repositories and **PPA equivalent** in **CentOS.**

sed -i -e '/^mirrorlist/d;/^#baseurl=/{s,^#,,;s,/mirror,/vault,;}' /etc/yum.repos.d/CentOS*.repo- This command updates the repository file and repairs it in a way.

Sudo yum clean all- Resets **YUM** in the server and looks for the current version of **YUM.**

We then write an official command to get a repository up and running within the CentOS server. The official command is….

Sudo yum install epel-release- This command installs an official repository to the CentOS server environment. This command helps install extra packages in the CentOS server environment.

```
[root@localhost yum.repos.d]# cd /etc/yum.repos.d/
[root@localhost yum.repos.d]# sudo sed -i -e '/^mirrorlist/d;/^#baseurl=/{s,^#,,;s,/mirror,/vault,;}
' /etc/yum.repos.d/CentOS*.repo
[root@localhost yum.repos.d]# sudo yum clean all
0 files removed
[root@localhost yum.repos.d]# sudo yum install epel-release
```

```
[root@localhost yum.repos.d]# cd /etc/yum.repos.d/
[root@localhost yum.repos.d]# sudo sed -i -e '/^mirrorlist/d;/^#baseurl=/{s,^#,,;s,/mirror,/vault,;}
' /etc/yum.repos.d/CentOS*.repo
[root@localhost yum.repos.d]# sudo yum clean all
0 files removed
[root@localhost yum.repos.d]# sudo yum install epel-release
CentOS Linux 8 - AppStream                                      5.7 MB/s | 8.4 MB     00:01
CentOS Linux 8 - BaseOS                                         4.0 MB/s | 4.6 MB     00:01
CentOS Linux 8 - Extras                                          20 kB/s |  10 kB     00:00
Dependencies resolved.
================================================================================================
 Package                    Architecture        Version                 Repository           Size
================================================================================================
Installing:
 epel-release               noarch              8-11.el8                extras               24 k

Transaction Summary
================================================================================================
Install  1 Package

Total download size: 24 k
Installed size: 35 k
Is this ok [y/N]: _
```

Two screenshots showing the **installation of extra packages** within the **CentOS server environment.**

```
================================================================================
Installing:
 epel-release          noarch          8-11.el8          extras          24 k

Transaction Summary
================================================================================
Install  1 Package

Total download size: 24 k
Installed size: 35 k
Is this ok [y/N]: y
Downloading Packages:
epel-release-8-11.el8.noarch.rpm                    97 kB/s |  24 kB     00:00
--------------------------------------------------------------------------------
Total                                               93 kB/s |  24 kB     00:00
CentOS Linux 8 - Extras                            681 kB/s | 1.6 kB     00:00
Importing GPG key 0x8483C65D:
 Userid     : "CentOS (CentOS Official Signing Key) <security@centos.org>"
 Fingerprint: 99DB 70FA E1D7 CE22 7FB6 4882 05B5 55B3 8483 C65D
 From       : /etc/pki/rpm-gpg/RPM-GPG-KEY-centosofficial
Is this ok [y/N]: y
Key imported successfully
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : epel-release-8-11.el8.noarch                           1/1
  Running scriptlet: epel-release-8-11.el8.noarch                           1/1
  Verifying        : epel-release-8-11.el8.noarch                           1/1

Installed:
  epel-release-8-11.el8.noarch

Complete!
[root@localhost yum.repos.d]# _
```

The final screenshot shows the installation of the first repository in the **CentOS server environment.**

CentOS does not use **PPA** (Personal Package Archives) ; the server environment instead uses External Repositories **(ER)** or third party repositories **(TPR). (TPR)** instead of **PPA.**

Ls /etc/yum.repos.d/

After writing this command we see a line that shows the different **TPR** within the directory for the server. The TPR installs were completed on 3/31/26.

Update command after PPA....

Sudo yum update- Updates the current packages within the repository for CentOS server.

```
Transaction Summary
================================================================================================
Upgrade  1 Package

Total download size: 25 k
Is this ok [y/N]: y
Downloading Packages:
epel-release-8-22.el8.noarch.rpm                                    45 kB/s |  25 kB     00:00
------------------------------------------------------------------------------------------------
Total                                                               24 kB/s |  25 kB     00:01
Extra Packages for Enterprise Linux 8 - x86_64                     1.6 MB/s | 1.6 kB     00:00
Importing GPG key 0x2F86D6A1:
 Userid     : "Fedora EPEL (8) <epel@fedoraproject.org>"
 Fingerprint: 94E2 79EB 8D8F 25B2 1810 ADF1 21EA 45AB 2F86 D6A1
 From       : /etc/pki/rpm-gpg/RPM-GPG-KEY-EPEL-8
Is this ok [y/N]: y
Key imported successfully
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                                        1/1
  Running scriptlet: epel-release-8-22.el8.noarch                                           1/1
  Upgrading        : epel-release-8-22.el8.noarch                                           1/2
  Running scriptlet: epel-release-8-22.el8.noarch                                           1/2
  Cleanup          : epel-release-8-11.el8.noarch                                           2/2
  Running scriptlet: epel-release-8-11.el8.noarch                                           2/2
  Verifying        : epel-release-8-22.el8.noarch                                           1/2
  Verifying        : epel-release-8-11.el8.noarch                                           2/2

Upgraded:
  epel-release-8-22.el8.noarch

Complete!
[root@localhost yum.repos.d]#
```

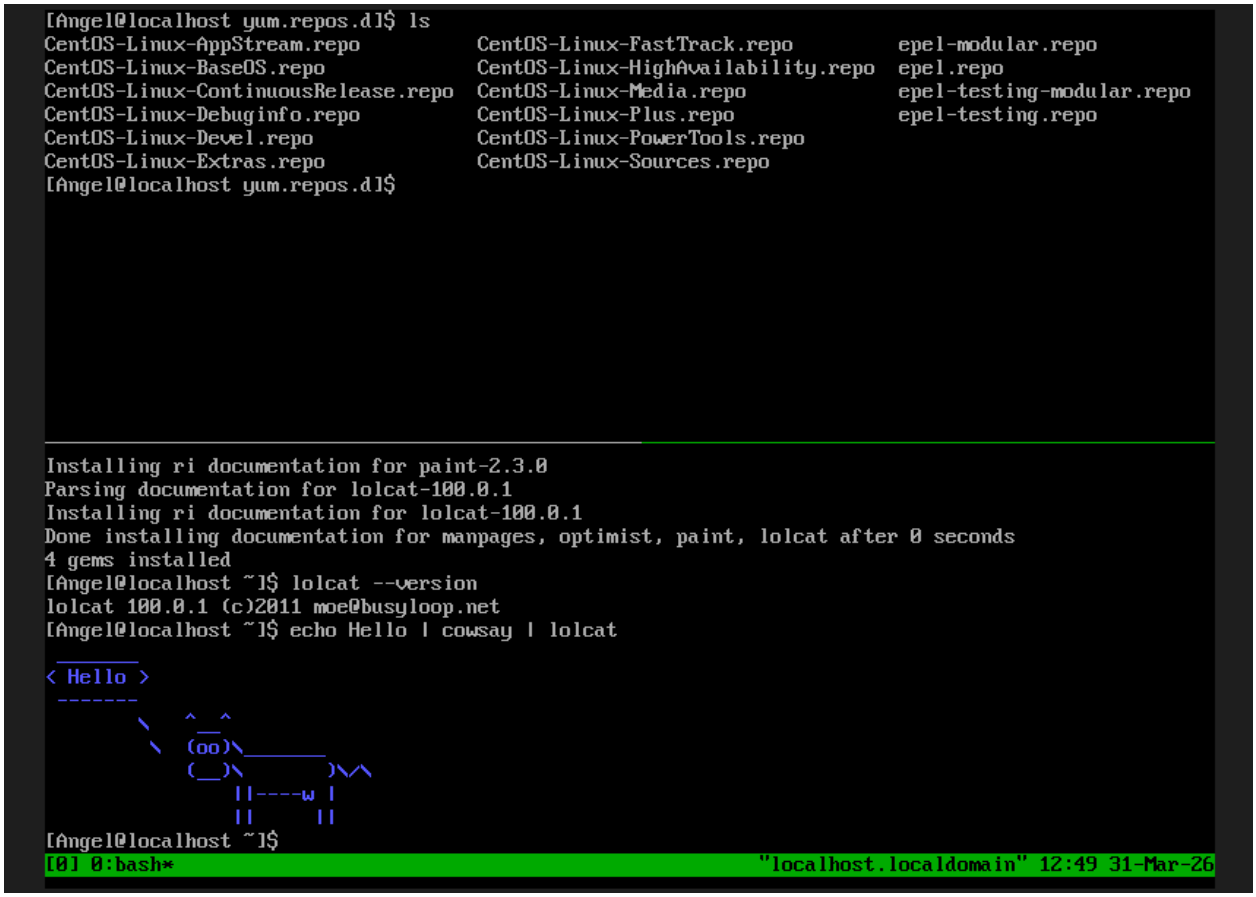
Output after running **sudo yum update** command. Written on **3/31/26.**

Sudo yum install tmux- Checks to see if tmux is installed. Tmux splits the terminal. Tmux was installed by default in CentOS as well. Installed on **3/31/26.**

```
[Angel@localhost ~]$ sudo yum install tmux
[sudo] password for Angel:
Last metadata expiration check: 20:58:35 ago on Mon 30 Mar 2026 03:13:28 PM EDT.
Package tmux-2.7-1.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[Angel@localhost ~]$ _
```

The steps to get into the tmux program, **tmux** is a program that splits windows into multiple terminals. The way I started the **tmux** program within the ubuntu server admin terminal was writing **tmux.** Then I did **ctrl b**, then **shift 5,** to split the screen vertically, these two commands open a new terminal screen within the server. To split the screen horizontally we would write **ctrl b,** then **shift + single quote (')** as if you were trying to write a **double quote (").**

To switch between the screens in horizontal split screen mode, we would write **ctrl b** then **ctrl up or ctrl down** on the arrow keys of your computer's keyboard.



To see the cowsay character and message animated we can write this command echo Hello | cowsay | lolcat.

To close the tmux program we write **Tmux kill-session,** this command ends the session that splits the two terminal screens. Written on **3/31/26.**

I wrote **sudo yum install snapd** command to install lolcat, along with other snap and snapd commands. The different commands had several issues and did not work as intended for lolcat and cowsay installation. I then tried installing lolcat using ruby, and had success using ruby and gem. Written on 3/31/26.

The command I wrote to install lolcat was **sudo yum install ruby,** I also wrote **gem install lolcat.**

```
[Angel@localhost ~]$ sudo yum install ruby
[sudo] password for Angel:
Last metadata expiration check: 21:37:29 ago on Mon 30 Mar 2026 03:13:28 PM EDT.
Package ruby-2.5.9-107.module_el8.4.0+847+ee687b6c.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[Angel@localhost ~]$ gem install lolcat
Successfully installed lolcat-100.0.1
Parsing documentation for lolcat-100.0.1
Done installing documentation for lolcat after 0 seconds
1 gem installed
[Angel@localhost ~]$

[0] 0:bash*                                    "localhost.localdomain" 12:51 31-Mar-26
```

Screenshot of using **sudo yum install ruby** and **gem install lolcat.**

```
[Angel@localhost ~]$ sudo yum install cowsay
Last metadata expiration check: 20:52:27 ago on Mon 30 Mar 2026 03:13:28 PM EDT.
Dependencies resolved.
================================================================================
 Package          Architecture     Version                  Repository      Size
================================================================================
Installing:
 cowsay           noarch           3.7.0-10.el8             epel            47 k

Transaction Summary
================================================================================
Install  1 Package

Total download size: 47 k
Installed size: 72 k
Is this ok [y/N]: y
Downloading Packages:
cowsay-3.7.0-10.el8.noarch.rpm                    127 kB/s |  47 kB     00:00
--------------------------------------------------------------------------------
Total                                              59 kB/s |  47 kB     00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :                                                        1/1
  Installing       : cowsay-3.7.0-10.el8.noarch                             1/1
  Running scriptlet: cowsay-3.7.0-10.el8.noarch                             1/1
  Verifying        : cowsay-3.7.0-10.el8.noarch                             1/1

Installed:
  cowsay-3.7.0-10.el8.noarch

Complete!
[Angel@localhost ~]$
```

Screenshot of Installing cowsay using the command **sudo yum install cowsay.**

Sudo yum install fail2ban- A command that installs fail2ban prevention software for the CentOS server.

```
[Angel@localhost ~]$ sudo yum install fail2ban
Last metadata expiration check: 0:07:03 ago on Tue 31 Mar 2026 01:12:35 PM EDT.
Package fail2ban-1.0.2-3.el8.noarch is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[Angel@localhost ~]$ _
```

Sudo yum install emacs- A command that installs the emacs text editor.

```
[Angel@localhost fail2ban]$ sudo yum install emacs
Last metadata expiration check: 0:11:36 ago on Tue 31 Mar 2026 01:12:35 PM EDT.
Package emacs-1:26.1-7.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[Angel@localhost fail2ban]$
```

cd /etc/fail2ban - This command shows that the **fail2ban** is its own directory. Remember to add a space in front of cat to ensure the command works, without the space there will be errors or complete command failure.

cd /etc/emacs- This command shows that **emacs** is its own directory, or does not work at all sometimes. Remember to add a space in front of cat to ensure the command works, without the space there will be errors or complete command failure.

When writing **ls** you can see the files that are within the directories **fail2ban** and **emacs.** Emacs does not contain a directory with files. Ran these commands on **3/31/26.**

sudo nano fail2ban.conf- Fail2ban.conf is one of the files in fail2ban directory with this command we can open the file to see its contents within the CentOS server. When opening the

file you can see the different operational default settings for **fail2ban,** like logging level and the different socket file paths.

```
  GNU nano 2.9.8                          fail2ban.conf

# Fail2Ban main configuration file
#
# Comments: use '#' for comment lines and ';' (following a space) for inline comments
#
# Changes:  in most of the cases you should not modify this
#           file, but provide customizations in fail2ban.local file, e.g.:
#
# [DEFAULT]
# loglevel = DEBUG
#

[DEFAULT]

# Option: loglevel
# Notes.: Set the log level output.
#         CRITICAL
#         ERROR
#         WARNING
#         NOTICE
#         INFO
#         DEBUG
# Values: [ LEVEL ]  Default: INFO
#
loglevel = INFO

# Option: logtarget
# Notes.: Set the log target. This could be a file, SYSTEMD-JOURNAL, SYSLOG, STDERR or STDOUT.
#         Only one log target can be specified.
#         If you change logtarget from the default value and you are
#         using logrotate -- also adjust or disable rotation in the
#         corresponding configuration file
#         (e.g. /etc/logrotate.d/fail2ban on Debian systems)
                                [ Read 92 lines ]
^G Get Help   ^O Write Out  ^W Where Is   ^K Cut Text   ^J Justify    ^C Cur Pos    M-U Undo
^X Exit       ^R Read File  ^\ Replace    ^U Uncut Text ^T To Spell   ^_ Go To Line M-E Redo
```

Fail2ban operational original settings in **Nano.**

sudo cp fail2ban.conf fail2ban.local- A command that makes a copy of the fail2ban.conf file and renames it to fail2ban.local. Writing **ls** we can see that the original file and the copied file are now within the same directory in **"fail2ban".**

When creating a **.local** copy of the file, I make sure the original service configurations remain the same and are not overwritten in any way when the fail2ban packages are updated by the YUM package manager. Writing **ls** we can see that the original file and the copied file are now within the same directory in **"fail2ban".**

```
[Angel@localhost fail2ban]$ sudo cp fail2ban.conf fail2ban.local
[Angel@localhost fail2ban]$ ls
action.d       fail2ban.d      fail2ban.lolcal  jail.conf   paths-common.conf
fail2ban.conf  fail2ban.local  filter.d         jail.d      paths-fedora.conf
[Angel@localhost fail2ban]$ _
```

Screenshot showing the original and copied file of **fail2ban.**

To get **fail2ban** to work we write **sudo systemctl enable fail2ban.service.** Then **sudo systemctl start fail2ban.service,** to start fail2ban.

Finally we write **sudo systemctl status fail2ban.service,** to see the current status of fail2ban and see if it is running properly.

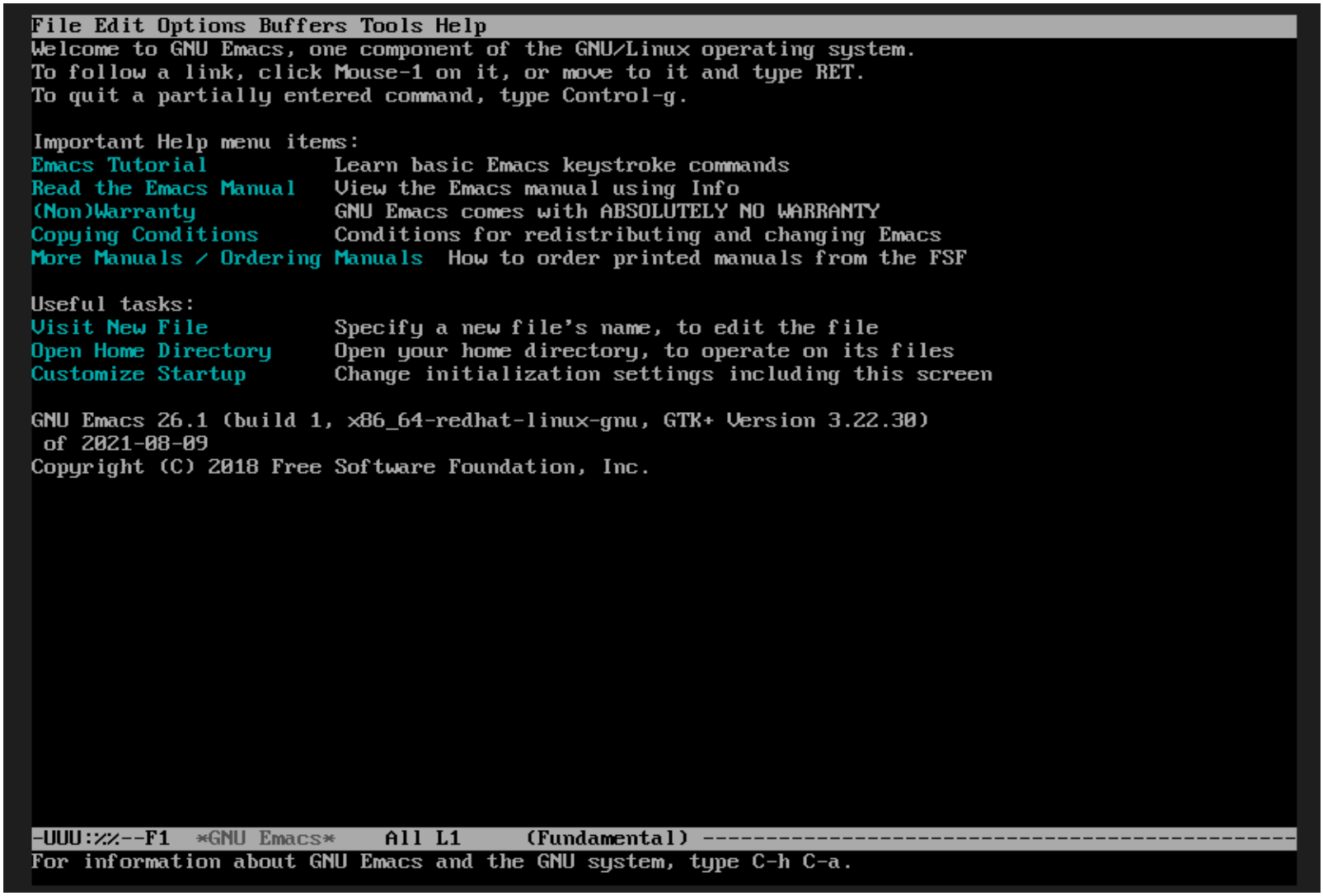
```
[Angel@localhost fail2ban]$ sudo systemctl enable fail2ban.service
Created symlink /etc/systemd/system/multi-user.target.wants/fail2ban.service → /usr/lib/systemd/syst
em/fail2ban.service.
[Angel@localhost fail2ban]$ sudo systemctl start fail2ban.service
[Angel@localhost fail2ban]$ sudo systemctl status fail2ban.service
● fail2ban.service - Fail2Ban Service
   Loaded: loaded (/usr/lib/systemd/system/fail2ban.service; enabled; vendor preset: disabled)
   Active: active (running) since Tue 2026-03-31 14:22:25 EDT; 17s ago
     Docs: man:fail2ban(1)
  Process: 2376 ExecStartPre=/bin/mkdir -p /run/fail2ban (code=exited, status=0/SUCCESS)
 Main PID: 2377 (fail2ban-server)
    Tasks: 3 (limit: 23669)
   Memory: 10.8M
   CGroup: /system.slice/fail2ban.service
           └─2377 /usr/bin/python3.6 -s /usr/bin/fail2ban-server -xf start

Mar 31 14:22:25 localhost.localdomain systemd[1]: Starting Fail2Ban Service...
Mar 31 14:22:25 localhost.localdomain systemd[1]: Started Fail2Ban Service.
Mar 31 14:22:25 localhost.localdomain fail2ban-server[2377]: Server ready
[Angel@localhost fail2ban]$
```

Above is a screenshot showing **fail2ban working** in the **CentOS** server environment.

To open emacs within the **CentOS** server I first changed the directory by writing **cd .. twice to change it back to home as the emacs directory is not similar to fail2ban that opens**

the same way. To open emacs, I wrote the command **emacs -nw**, which is emacs space No Window. No window works within the terminal space, instead of a separate graphics window.



Screenshot of **Emacs** running in the **CentOS** server environment.

To close the emacs new window you would press **ctrl x and then ctrl c,** this is very important to do as if you do not do this you will be stuck in the emacs welcome screen and would have to restart your server. Written **3/31/26.**

Echo Hello | **cowsay** | **lolcat-** This is a command that prints text for the cowsay character to say. With lolcat there are different colors that are displayed for the character and text within the CentOS server environment. The lolcat character being in color works for most servers but for Ubuntu it does not always work.

To see the cowsay character be colorful and animated we would write this command **Echo Hello | cowsay | lolcat-a**

```
[Angel@localhost /]$ echo Hello | cowsay | lolcat -a
 _______
< Hello >
 -------
        \   ^__^
         \  (oo)\_______
            (__)\       )\/\
                ||----w |
                ||     ||
[Angel@localhost /]$ _
```

usr/share/cowsay/cows- use - F-. - a command to see different character options.

When combining different ideas and commands from Ubuntu for **CentOS** I came up with a different type of command that gets the tux linux character to be **animated** and also say something.

The command I had written for this to work was, **echo "hey" | cowsay -f tux | lolcat -a.**

```
[Angel@localhost /]$ echo "hey" | cowsay -f tux | lolcat -a
 _____
< hey >
 -----
   \
    \
        .--.
       |o_o |
       |:_/ |
      //   \ \
     (|     | )
    /'\_   _/`\
    \___)=(___/

[Angel@localhost /]$ _
```

Tux and **Cow** commands with colors and animations written on **3/31/26.**

Change vim to vi. Written on **4/1/26.**

Cd ~ - Changes the current directory.

Nano ~/.bashrc- Opens a nano file in the CentOS server environment.

A colorful text file is then opened, at the end of the file we write, **alias vi=vim** to change **vi to vim** in the terminal.

```
  GNU nano 2.9.8                    /home/Angel/.bashrc                    Modified

# .bashrc

# Source global definitions
if [ -f /etc/bashrc ]; then
        . /etc/bashrc
fi

# User specific environment
if ! [[ "$PATH" =~ "$HOME/.local/bin:$HOME/bin:" ]]
then
    PATH="$HOME/.local/bin:$HOME/bin:$PATH"
fi
export PATH

# Uncomment the following line if you don't like systemctl's auto-paging feature:
# export SYSTEMD_PAGER=

# User specific aliases and functions
alias vi=vim_

^G Get Help   ^O Write Out  ^W Where Is   ^K Cut Text   ^J Justify    ^C Cur Pos    M-U Undo
^X Exit       ^R Read File  ^\ Replace    ^U Uncut Text ^T To Spell   ^_ Go To Line M-E Redo
```

We then write **ctrl O** to save/write out, then **press enter then ctrl x** to close the file.

```
~
~
~
~
~
~
~
~
~
~
~
~
~                    VIM - Vi IMproved
~
~                     version 8.0.1763
~                  by Bram Moolenaar et al.
~            Modified by <bugzilla@redhat.com>
~         Vim is open source and freely distributable
~
~                  Sponsor Vim development!
~        type  :help sponsor<Enter>    for information
~
~        type  :q<Enter>               to exit
~        type  :help<Enter>  or  <F1>  for on-line help
~        type  :help version8<Enter>   for version info
~
~
~
~
~
~
~
~
~
~
~
~
~
```

CentOS server **Sources:**

https://www.cyberciti.biz/faq/rhel-centos-fedora-linux-yum-command-howto/ **Important CentOS commands.**

https://jms1.info/linux/centos7-vault.html **CentOS Repository commands.**

https://serverspace.us/support/help/configure-repositories-on-centos-8 **Extra Packages for Enterprise Linux command, along with other examples for CentOS server.**

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https://snapcraft.io/install/lolcat/centos **lolcat command error commands. Originally used for lolcat.**

https://linuxize.com/post/how-to-install-ruby-on-centos-7/ **Installing ruby.**

https://www.scribd.com/document/803731589/4-Module-3-Installing-and-Exploring-cowsay-on-Ubuntu **Steps on how to install cowsay and examples. Question 3 source. Changed a little for CentOS.**

https://www.tecmint.com/lolcat-color-output-linux-terminal/ **How to install lolcat with command examples and screenshots. Question 3 source. Changed for CentOS, changed apt to yum, a lot of the apt commands are similar to yum.**

https://help.ubuntu.com/community/Fail2ban **Fail2ban configuration, install commands. Changed for CentOS, changed apt to yum, a lot of the apt commands are similar to yum.**

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https://www.digitalocean.com/community/tutorials/how-to-use-the-emacs-editor-in-linux **How to install emacs. Changed for CentOS, changed apt to yum, a lot of the apt commands are similar to yum.**

https://superuser.com/questions/203298/ubuntu-keep-emacs-in-the-terminal **How to enter the emacs screen. Changed for CentOS, changed apt to yum, a lot of the apt commands are similar to yum.**

https://web.physics.ucsb.edu/~pcs/apps/editors/emacs-basics.html **How to exit emacs and other commands. Changed for CentOS, changed apt to yum, a lot of the apt commands are similar to yum.**

https://linuxize.com/post/getting-started-with-tmux/ **Tmux example commands. Changed for CentOS, changed apt to yum, a lot of the apt commands are similar to yum.**

Tutoring Assistance.